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Dear Editor-in-Chief,

We submit our manuscript, "Resource Frames, Graph Scope, and Rail States: Conditional Decision Boundaries for Reserve-Force Bus versus Rail–Bus Multimodal Transport," for consideration as a regular article in IEEE Access.

Reserve-force mobilization planning frequently reduces a wartime transport choice to "bus versus rail–bus." Our central finding is that this comparison is uninterpretable unless five design choices are reported together: how road and rail resources are counted, how much of the national road graph is retained, whether rail is modeled as an available/degraded/unavailable service state, how the multimodal road fleet is internally allocated, and which response policy is allowed. We make each choice an explicit experimental object on a public-data Korean corridor (Songpa–Goseong, ∼1,000 personnel, 24-h window) simulated over 16 campaign sets (32,790 runs) with common-random-number pairing, and we show that reasonable frame and allocation choices can reverse, inflate, or dissolve the apparent modal winner. The headline result is a conditional decision boundary: under an equal-road-fleet frame with normally available rail, the long-haul road-damage break-even is a band (1.150–1.277 for well-split fleets; 1.742 for the arbitrary legacy split), whose endpoints reproduce on the full 360,556-node reference graph and whose allocation optimum survives nested-selection and independent-holdout correction.

The contribution is deliberately narrow: an evaluation-and-reporting protocol assembled from established methods (CRN pairing, bootstrap and joint-seed crossing inference with reported valid-replicate counts, crossed two-way threat/arrival resampling, Morris screening, full-graph scope auditing), demonstrated on two related corridors. All inputs are public-data-derived and carry explicit source classes; no claim of calibration, prediction, or field readiness is made.

All code, input tables, campaign outputs, and a manuscript-table-to-CSV map are openly archived (Zenodo DOI 10.5281/zenodo.23187709, MIT license); the manuscript is reproducible from the pinned environment documented in its reproducibility section.

The manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the submission and declare no conflicts of interest.

We thank you for your consideration.

Sincerely,
Hyunho Kim, on behalf of all authors